

General Industries Hub

Enterprise Inventory Management

Store Keeper Manual

Site-level material handling, daily entries, returnables.

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Confidential - for authorized personnel only

GI Hub - Role booklet - Store Keeper Manual

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1. Introduction & System Overview

1.1 What the system does

The General Industries (GI) Hub is a multi-site warehouse inventory, procurement and asset-management system. It tracks every material movement - issues, receipts, returns and corrections - through a two-stage approval process, and gives you live stock figures, oldest-stock-first discipline on anything with an expiry date, stock valuation, serialised tool and equipment tracking, and a complete audit trail across every site you manage.

It runs in a web browser on a phone, a tablet or a desktop, and is also available as an installable app for Windows, macOS and Android.

1.2 Core principles

Principle	What it means
Stock is calculated, never stored	Stock on hand is always received, minus consumed, minus returned, recalculated from the movement history every time it is shown. No running total is kept anywhere, so no running total can drift out of step with reality.
Two-stage approval	Store Keepers stage entries -> HOD reviews and commits at End-of-Day. Nothing touches the permanent ledger without HOD approval.
Site isolation	HODs and Supervisors see only their own site's stock. Only Admin sees all sites. Cross-site moves require formal request + approval.
Audit-first	Every consequential action is recorded with the user's name, the exact time and what changed. Even a deletion leaves a trace.
Nothing to maintain	Upgrades apply themselves when the system starts. Nobody has to run a database update by hand.
Procurement chain	One continuous workflow from a site's request through to material on the shelf. Logistics owns purchase orders; Warehouse owns physical receiving and delivery notes; the Site HOD approves what is being sent; the Store Keeper confirms it arrived. Every handover is recorded.
RL/BL strict separation	Rubber Lining and Brick Lining items NEVER share a PO line group, a DN, or a warehouse aggregation. The system rejects mixed-family DNs by design and tags each line with its family on insert.

Principle	What it means
Warehouse-blind pricing	Warehouse users can see materials and quantities but NEVER see Unit_Price, Total_Price, or any monetary header field on a PO. Three independent enforcement layers guarantee this.

1.3 The transaction lifecycle (the heart of the system)

Every stock movement travels the same four steps. Nothing reaches the permanent record until a Head of Department has approved it.

Step	Who does it	What happens	Where it sits afterwards
1	Store Keeper	Enters the issue, receipt or return on the Entry Log	Staged, awaiting approval
2	Head of Department	Reviews it at End-of-Day and may edit the quantity, approve, or reject with a reason	Still staged until they decide
3	System	Commits the approved entry to the permanent stock record	Permanent - stock figures move now
4	System	Sends the WhatsApp and in-app notifications, and writes the audit entry	Permanent audit history

Receipts, issues, returns and stock adjustments all follow this same shape. A rejected entry never reaches step 3, so it never affects a stock figure - but the rejection itself is still recorded, together with the reason.

1.3a The procurement chain

Buying material is a relay. Each role does one thing and hands over to the next, and every handover appears in the receiving role's own queue.

Step	Role	Action	Lands in
1	Site HOD	Submits a Purchase Request	Logistics - Incoming PRs
2	Logistics	Issues the Purchase Order	Site HOD is notified; Admin sees it for oversight
3	Logistics	Assigns the order (or individual lines) to a warehouse	Warehouse - Incoming Assignments
4	Warehouse	Acknowledges, then	Warehouse - Receiving

Step	Role	Action	Lands in
		records the physical arrival from the vendor	
5	Warehouse	Prepares the Delivery Note	Logistics - DN approval queue
6	Logistics	Approves the delivery date	Site HOD - DN Approvals
7	Site HOD	Approves the contents	Store Keeper - Incoming DNs
8	Store Keeper	Confirms physical arrival on site	Permanent stock record; the delivery closes

Three side-paths exist for when things do not go to plan:

- Vendor returns - any role can raise one. The order line reopens so the material can be re-delivered.
- Reschedules - Warehouse or Site HOD request a new date; Logistics decides.
- Force-closure - Logistics only, for an order that will never complete. Both Admin and the Site HOD are notified automatically.

A Delivery Note may never mix Rubber Lining and Brick Lining material. The system refuses to create one, rather than warning about it.

1.4 Currency, dates, units

- Currency: Saudi Riyal (SAR) throughout.
- Dates: shown as DD/MM/YYYY or DD MMM YYYY.
- Time zone: the site's local time. Audit entries record the time to the second.
- Units of measure: each item carries its own unit - pieces, boxes, rolls, cans, metres, kilograms. There is no automatic conversion: an item is issued in the same unit it was received in. If a material arrives in boxes and is used in pieces, record the conversion in the item's description and keep one unit for both directions.

2. Roles, Permissions & Page Access

2.1 Role hierarchy

Roles are ranked, but the ranking is not a straight line - two pairs sit side by side at the same level and differ only in what they are scoped to.

Level	Role	Scoped to
0	Store Keeper	One site
1	Warehouse User	One warehouse
1	Supervisor	One site
1	Quality Control (QC)	One site or one warehouse - never both
2	Head of Department	One site
2	Head of Qualities (QC-HOD)	All sites, Surface Shield only, read + escalate
3	Logistics	All sites
3	Auditor	All sites, read-only
4	Admin	Everything

The Head of Qualities is level 2 but reads across every site, and that is not a contradiction. Cross-site reach normally comes from being level 3 - and level 3 would also have handed this role every endpoint the rank admits, which is most of the system. Instead it is level 2 with a named cross-site exemption, and the rank grants it nothing at all: a level check refuses it outright, so it reaches only what names the role explicitly. Its whole surface is the Quality Oversight page. See §23.

The QC role is scoped on one of two axes, and exactly one. Material is inspected either where it arrives (a warehouse) or where it lands (a site), and which one depends on how your operation is organised. A QC account is therefore created against a site or against a warehouse, and an account carrying neither - or both - sees nothing at all rather than everything. That is deliberate: an inspector with an unclear remit is a bigger problem than an inspector with none.

2.2 Page access matrix - which role can open which page

[!] A higher rank does not open a lower role's workspace. The Logistics Portal, Warehouse Portal, Entry Log, HOD Portal, Supervisor Portal, Material Estimator and Man-Hours pages are locked to their own role exactly - Logistics outranks a Head of Department but still cannot open the HOD Portal. Read that caveat together with the table below it, not instead of it: the caveat says a page is locked to "its own role", and the table is what names that role. A Head of Department can open Man-Hours and the Material Estimator; those two are locked to the HOD, not away from them. Admin is the single exception and reaches every workspace deliberately, for support.

Page	Store Keeper	Warehouse User	Supervisor	QC	HOD	Logistics	QC-HOD	Auditor	Admin
Live Dashboard	[OK]	[OK]	[OK]	[X]	[OK]	[OK]	[X]	[OK]	[OK]
Stock	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]	[X]	[OK]	[OK]
Rack Locator	[OK]	[OK]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK]
Assets	[OK]	[OK]	[OK]	[X]	[OK]	[OK]	[X]	[OK]	[OK]
Entry Log (all tabs)	[OK]	[X]	[X]	[X]	[X]	[X]	[X]	[X]	[OK] (shadow)
Supervisor Portal	[X]	[X]	[OK]	[X]	[X]	[X]	[X]	[X]	[OK] (shadow)
HOD Portal -> Approvals	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]
HOD Portal -> Executive Summary, Burn Rate, Documents, Low Stock, PRs, Cross-Site	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK] (read)	[OK]
Lining Coverage	[X]	[X]	[X]	[X]	[OK]	[OK]	[X]	[OK]	[OK]
Bulk Excel Import	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]
WBS & Work	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]

Page	Store Keeper	Warehouse User	Supervisor	QC	HOD	Logistics	QC-HOD	Auditor	Admin
Types									
Material Estimator (SME)	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK] (read)	[OK]
Man-Hours / Manpower Tracking	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]
Execution Entries	[OK]	[X]	[OK]	[X]	[OK]	[X]	[X]	[X]	[OK]
Reports	[X]	[X]	[X]	[X]	[OK]	[OK]	[X]	[OK] (read + download)	[OK]
Logistics Portal (Procurement)	[X]	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK] (shadow)
Warehouse Portal (Receiving & DN)	[X]	[OK]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK] (shadow)
Quality -> Inspections	[OK] (read)	[OK] (read)	[X]	[OK] (decides)	[OK] (read)	[OK] (read)	[X]	[OK] (read)	[OK]
Quality -> QC Accounts	[X]	[OK]	[X]	[X]	[OK]	[OK]	[X]	[X]	[OK]
Quality Oversight (\$23)	[X]	[X]	[X]	[X]	[X]	[X]	[OK]	[X]	[OK]
Safety &	[OK]	[X]	[X]	[X]	[OK]	[OK]	[X]	[X]	[OK]

Page	Store Keeper	Warehouse User	Supervisor	QC	HOD	Logistics	QC-HOD	Auditor	Admin
People -> PPE Forecast									
Safety & People -> PPE Usable Time	[OK]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]
Safety & People -> Employees	[OK]	[X]	[OK]	[X]	[OK]	[X]	[X]	[OK]	[OK]
Records (Inventory, ledgers, POs, PRs)	[X]	[X]	[X]	[X]	[X]	[OK]	[X]	[OK] (read)	[OK]
Valuation & 30-Day Burn (\$24.3)	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK]	[OK]
Admin Portal	[X]	[X]	[X]	[X]	[X]	[X]	[X]	[X]	[OK]
Documents - Security - Training - Feedback	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]

Where each role lands after signing in: Store Keeper -> Entry Log -> Issue Stock - Warehouse User -> Warehouse Portal - Supervisor -> Supervisor Portal - QC -> Quality -> Inspections - HOD -> HOD Portal -> Approvals - Logistics -> Logistics Portal - Head of Qualities -> Quality Oversight - Auditor ->

Dashboard - Admin -> Admin Portal -> Console.

2.3 What each role can do

One list per role, in plain words. If you only want to know whether your account can do a thing, read your own entry - each is complete on its own.

2.3.1 Store Keeper - what a Store Keeper can and cannot do

Can: open the Entry Log and post every stock movement - receive, issue, return, adjust, count, returnable items, OCR import; see Incoming Deliveries and Supervisor Requests; open the Dashboard, Stock, Rack Locator, Assets, PPE Forecast, PPE Usable Time, Employees and Execution Entries; read quality inspections; download documents.

Cannot: approve anything (an HOD approves what a Store Keeper stages); open the HOD, Supervisor, Logistics or Warehouse portals; open Reports, Records, the Material Estimator or Man-Hours; see another site's stock.

Scope: one site.

2.3.2 Supervisor - what a Supervisor can and cannot do

Can: raise material requests from the Supervisor Portal; post Execution Entries; open the Dashboard, Stock, Assets and Employees.

Cannot: post stock movements (the Entry Log is the Store Keeper's); approve requests; open Reports, Records, the Material Estimator, Man-Hours or any other role's portal.

Scope: one site.

2.3.3 Quality Control (QC) - what an inspector can and cannot do

Can: decide inspections - pass, fail or hold - from Quality -> Inspections; open the Stock page to see what is waiting and what their decision released.

Cannot: open the Dashboard, post any stock movement, approve a purchase, create QC accounts, or move themselves between sites (that is an Admin decision). See §22.

Scope: exactly one site or exactly one warehouse - never both, never neither.

2.3.4 Head of Department (HOD) - what an HOD can and cannot do

Can - and this is the question people ask most: open the Man-Hours / Manpower Tracking page and the Material Estimator. Both are exact-locked to the HOD (and Admin for support); the lock is what admits an HOD, not what excludes them. That includes all thirteen Man-Hours tabs - roster, timesheets, estimator, variance, scorecard, employee-wise, the four execution views, the Manpower Planner and the SME Session Plan (§19).

Also: approve or reject everything the site stages (receipts, issues, returns, adjustments, counts); create and submit Purchase Requisitions to Logistics; raise Cross-Site Requests; run Reports; open the

Executive Summary, Burn Rate, Low Stock, Lining Coverage, Document Library and Bulk Excel Import; create QC accounts for their site; post Execution Entries; open PPE Forecast, PPE Usable Time and Employees.

Cannot: open the Logistics Portal, the Warehouse Portal, the Supervisor Portal, the Entry Log, Quality Oversight or the Admin Portal; raise a Purchase Order (Logistics does that); see another site's stock - though they may request material from one.

Scope: one site.

2.3.5 Head of Qualities (QC-HOD) - what this role can and cannot do

Can: open the Quality Oversight page and nothing else. Across every site and warehouse, but only for Surface Shield material: the controlled category is filtered in the database on every query, so it is the boundary of the role rather than a filter on a page. Seven views - overview, Surface Shield purchase orders, the MTC register, where it is used, stagnation and expiry, escalations, and the 90/60-day thresholds. They may raise and close escalations and retune those two thresholds.

Cannot: approve or reject a quality inspection; move, issue or receive stock; raise a PR or PO; open any other portal or page; see PPE, tools, consumables or any other category. See §23.

Scope: all sites and all warehouses; Surface Shield only.

2.3.6 Logistics Coordinator - what Logistics can and cannot do

Can: run the whole procurement chain from the Logistics Portal - receive submitted PRs, raise Purchase Orders, assign POs to warehouses, manage vendors, track DN status; open Records (inventory, ledgers, POs, PRs), Reports, the Warehouse Portal, Lining Coverage, the Rack Locator, PPE Forecast, and quality inspections; create QC accounts.

Cannot: open the HOD Portal, the Entry Log, the Supervisor Portal, the Material Estimator, Man-Hours or the Admin Portal - outranking a role does not open its workspace.

Scope: all sites, no site lock.

2.3.7 Warehouse Operator - what a Warehouse User can and cannot do

Can: receive against POs, prepare and dispatch Delivery Notes from the Warehouse Portal; open the Dashboard, Stock, Rack Locator and Assets; read quality inspections; create QC accounts for their warehouse.

Cannot: open the Logistics Portal, the HOD Portal, the Entry Log, Reports, Records, the Material Estimator or Man-Hours.

Scope: one warehouse.

2.3.8 Auditor - what a view-only Auditor can and cannot do

Can: read across every site - Dashboard, Stock, Assets, Records, Reports (including downloads), Lining Coverage, the Material Estimator, the HOD Portal's read-only views (Executive Summary, Burn Rate, Documents, Low Stock, PRs, Cross-Site Requests), Employees and quality inspections.

Cannot: change anything at all, anywhere. Every write route is refused by the server, not merely hidden. Not Man-Hours, not the Entry Log, not any approval. See §20.

Scope: all sites, read-only.

2.3.9 Administrator - what an Admin can do

Everything, including every other role's workspace as a support shadow, plus the Admin Portal: users, access requests, overdue actions, inventory master, the audit log and the console.

2.4 Site scope by role

Role	What they see
Store Keeper	Their own site only - they cannot view another site's stock.
Warehouse User	Their own warehouse only - POs assigned to them, DNs they've prepared, items received from vendors. Tied to users.Warehouse_ID.
Supervisor	Their own site only - Reports, Live Dashboard, Burn Rate all site-locked.
Quality Control (QC)	Whichever single place they were created against: one site, or one warehouse. A site QC sees inspections raised at that site and has no warehouse business at all; a warehouse QC is pinned to its warehouse exactly as a Warehouse User is. An account with no binding, or with both, sees an empty list - it never falls back to seeing everything. Moving a QC to another site is a request, decided by an Admin, not something the QC or their HOD can do alone.
HOD	Their own site only - but they can REQUEST material from other sites (Cross-Site tab) and submit PRs to Logistics.
Logistics	All sites globally for PRs and POs they manage. No site lock - they sit above the site boundary.
Head of Qualities	All sites globally, but ONLY Surface Shield material - every query on their page is filtered to the controlled category in the database, so the category is the boundary of the role rather than a filter on a page. They read, and they send escalations; they cannot approve an inspection, move stock, raise a PR or open any other portal. See §23.
Auditor	All sites globally, read-only. Sits at level 3 so it is not site-locked - an auditor pinned to one site could not audit. It can open the Dashboard, Stock, Records, Reports and Lining Coverage and change nothing anywhere. See §20.

Role	What they see
Admin	All sites + all warehouses globally - has the "All Sites" filter on every multi-site view; warehouse picker in sidebar when shadowing the Warehouse Portal.

2.5 Default seeded accounts

The first time the app starts, these accounts are created. Change the passwords immediately.

Username	Password	Role
admin	admin2026	Admin
hod	hod2026	HOD
supervisor	super2026	Supervisor
worker	floor2026	Store Keeper

No default logistics, warehouse_user, qc, qc_hod or auditor accounts are seeded. All of them are strictly admin-created - go to Admin Portal -> Users -> Add user. This is intentional: the procurement chain has commercial visibility (Logistics sees prices, Warehouse routes inventory) and seeded credentials would be a security liability. When you create a warehouse_user, set their Warehouse_ID to one of the values from Admin Portal -> Master DB Editor -> warehouses table - without it, the user lands on the Warehouse Portal and sees an error telling them to ask Admin.

3. Login, Sidebar & Common Elements

3.1 Login screen

Elements:

- Live | Practice switch - above the username. Live is the real system and is selected unless you choose otherwise. Practice is a separate training copy with made-up data, where nothing you do affects Live. Switching signs you out. See §26 Practice Mode for everything about it.
- Username text box - your assigned username (case-sensitive).
- Password text box - masked input. Min 8 characters policy enforced on creation.
- Sign In button - checks your credentials and takes you straight to the first page your role can open.
- "Don't have an account? Request access" link - opens a self-service registration form.

Registration form elements:

- Username - must be unique
- Password / Confirm Password
- Role requested - selectbox: Store Keeper / Supervisor / HOD (Admin role cannot be self-requested)
- Site_ID - which site to associate the account with
- Phone Number - used for WhatsApp alerts (format: +966 5X XXX XXXX)
- Submit Request button - creates a pending_users row + audit log + WhatsApp alert to all admins

Admin must approve via Admin Portal -> Users tab before the user can log in.

3.2 Sidebar (visible after login)

Every page shares this sidebar. Reading top to bottom:

Element	Purpose	Notes
GI Hub bolt icon + version	Branding	Shows v3.0.0
Role card	Your username + role badge	Color-coded: grey=Store Keeper, emerald=Warehouse, blue=Supervisor, indigo=HOD, sky=Logistics, gold=Admin
Notifications bell (NEW v3.0)	Unread count + inbox dialog	Red badge if N>0, primary button changes to "Open inbox (N unread)". Modal shows recent procurement events with mark-read controls. See §3.6
"Navigate to:" radio	Page picker	Only shows pages allowed by your role

Element	Purpose	Notes
INVENTORY ALERTS section	Compact stock-alert badge	Visible to Supervisor/HOD/Admin only. Shows count of items below minimum (or "All levels adequate" if clear)
Bug/Feature reporting	Opens a dialog	See §3.4
Theme toggle	Dark / Light mode switch	Persists per-session
Sign Out button	Ends the session	Audit-logs the LOGOUT event

3.3 Brand header (top of every page)

Variant	Used by	Visual
Standard header	Live Dashboard, Entry Log	Gold subtitle accent + current date
HOD header	HOD Portal	Purple subtitle accent + current date
Admin header	Admin Portal	Gold accent + green/amber pulse chip ("All systems operational" / "Degraded") + current date

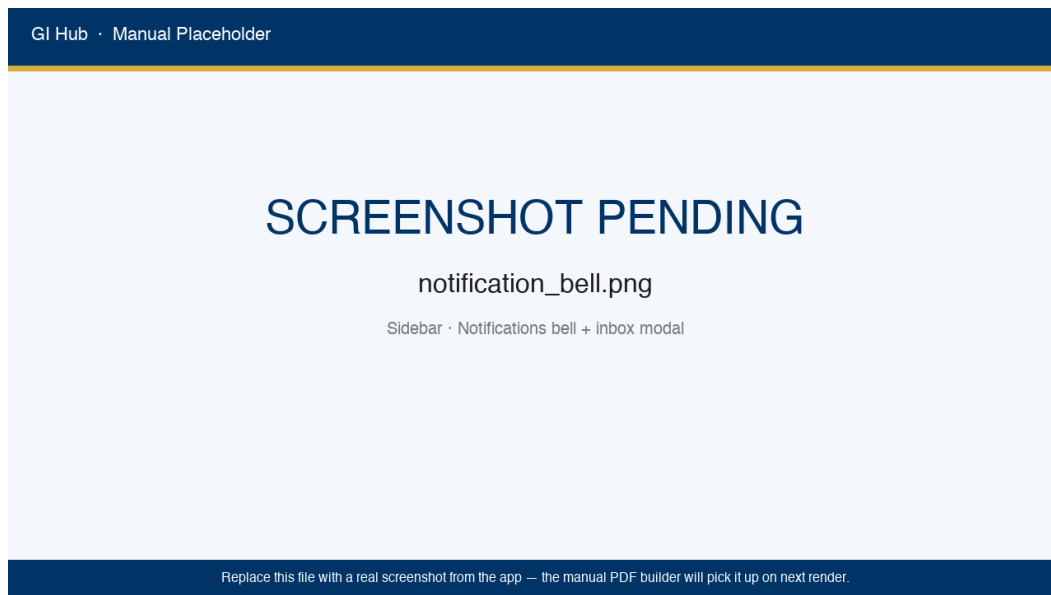
3.4 Bug / Feature reporting dialog

Available to every user from the sidebar:

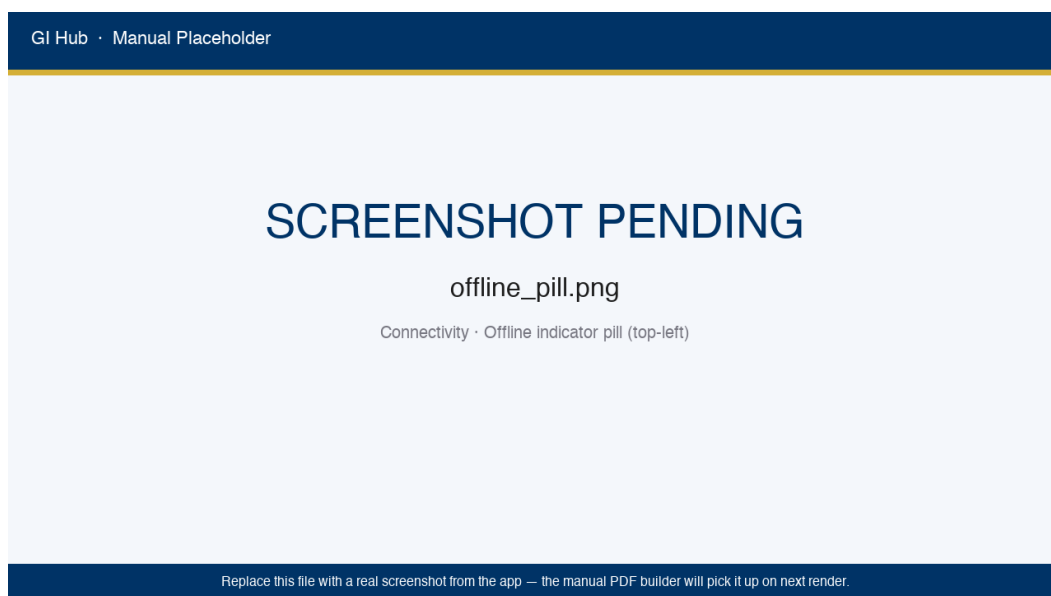
- Type selectbox - Bug Report / Feature Request
- Page dropdown - which page the issue/idea relates to (Live Dashboard, Entry Log, HOD Portal, Admin Portal, Reports, Other)
- Description textarea - up to 200 characters
- Submit button - writes to bug_reports table with user, timestamp, type, page, description. Visible to Admin in the Admin Portal Reports & Bugs tab for triage.

3.6 Notifications bell (NEW in v3.0)

Sits between the role card and the navigation radio. Every signed-in user sees their own personalised inbox of procurement-chain events.



Sidebar - Notifications bell with unread badge + inbox modal



Connectivity - offline indicator pill appears top-left when the browser loses network (Phase 7E)

The button:

- No unread: "Open inbox" (secondary button, no badge)
- 1+ unread: "Open inbox (N unread)" (primary button, red pill badge with the count, capped at "99+")

The inbox dialog (click the button):

- Only unread toggle - on by default. Flip off to see your full history.
- Per-notification card - colour-coded left border by severity:
- Critical (red) - force-closures, T-0 delivery reminders
- Warning (amber) - T-1/T-2 reminders, reschedule requests
- Success (green) - DN approved / received successfully
- Info (blue) - PR submitted, PO issued, assignments
- Per-row Mark read - flips just that notification

- Bulk [OK] Mark all as read - flips every visible row (respecting the role + site + warehouse scope you'd see normally)

What gets sent here:

Event you'll see	Triggered by
New PR from a site (Logistics only)	Site HOD pressing " Submit PR(s) to Logistics"
PO issued (Site HOD)	Logistics creating a PO against your site's PR
PO assigned to warehouse (Warehouse only)	Logistics routing a PO to your WH
DN awaiting your approval (HOD)	Logistics approving a DN delivery date
Incoming DN ready to receive (SK)	HOD approving DN content
Reschedule requested (Logistics)	Warehouse or HOD asking to push a date
Reschedule decided (requester)	Logistics approve/reject
Force-closure (Admin + originating HOD)	Logistics force-closing a PR/PO/line
Vendor return raised (Logistics)	Any role raising a return
Delivery reminder T-2 / T-1 / T-0	The daily sweep job (see §9)

The bell is tolerant - if the notification lookup fails, the badge silently shows 0 instead of breaking the sidebar. In-app notifications ALWAYS fire; the WhatsApp side is gated by the per-event WhatsApp toggles in the system settings (see §9).

3.5 Overdue Returnable banner (Store Keepers only)

When a Store Keeper logs in and there are overdue returnable items at their site, a red banner appears at the top of any page they navigate to:

> [!] OVERDUE ITEMS - Action Required: N borrowed item(s) past expected return: <item names>. Go to the Returnable Items tab to follow up.

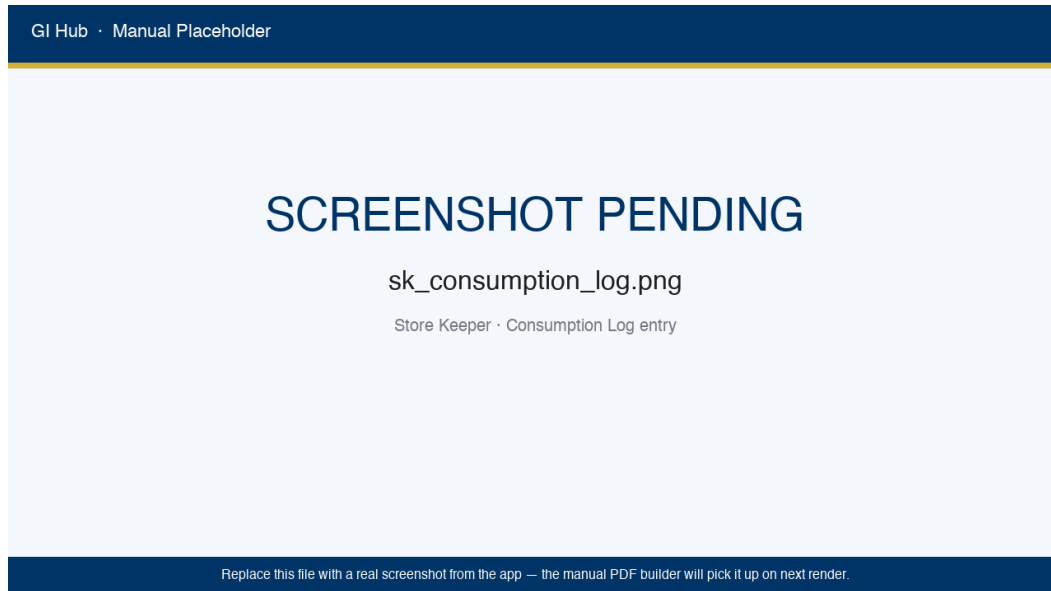
3.7 Saving an entry when the signal drops

If the network drops while you save a receipt, issue, return, stock adjustment or bulk batch, GI Hub keeps the entry on your device and shows a saved offline message instead of an error. A badge in the header shows how many entries are waiting. They are sent by themselves when the connection returns, and you can press the badge to send them straight away.

An entry is never saved twice. Each save carries its own reference, and the server recognises it if it arrives again. That can happen when the page reloads while the entry is being sent, or when the connection drops just as the server confirms. The second arrival is matched to the first, and only one entry reaches your HOD. You never need to check for, or delete, a duplicate caused by a bad connection.

4. Store Keeper Manual

The Store Keeper is the warehouse-floor operator. They see only the Entry Log page and the sidebar shell.



Store Keeper Consumption Log entry - main daily workflow

4.1 Pages visible

- Entry Log
- Locator - find which rack a material is on, or scan a rack to see what should be on it (see §21.8)
- Assets - the serialised tools and equipment register, with its own record and location history for each physical item (see §21.7)

The Locator and Assets pages are available to every role, Store Keeper included. They are deliberately placed at the top of the menu rather than buried inside a portal, because the people who need them are usually standing in the warehouse holding a scanner.

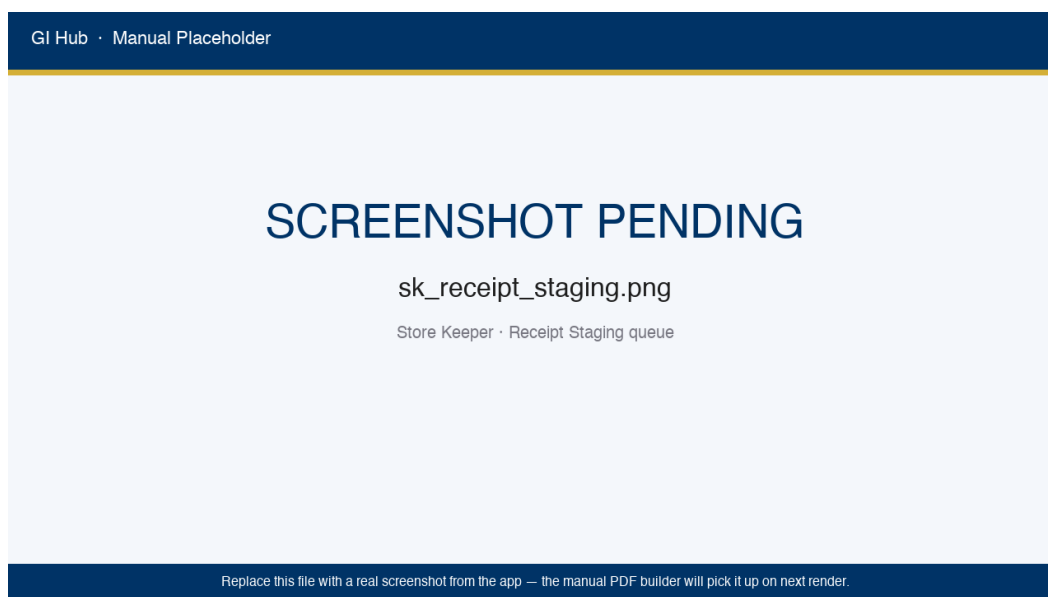
4.2 Entry Log - Tab structure

The Entry Log has four tabs:

1. Consumption Log - record material issued out
2. Receipt Staging - record material arriving in
3. Returnable Items - track tools/items temporarily issued
4. Stock Count - submit physical-count reconciliations

4.3 Entry Log -> Consumption Log

This is where every material consumed by site operations is recorded.



Receipt Staging queue - review incoming materials before submitting to HOD



Supervisor Requests tab - approve / reject worker material requests

4.3.1 Top section: Bulk OCR upload (expander)

Upload Handwritten Consumption List (OCR) - for when you have a handwritten list to bulk-stage instead of typing each row.

Element	Purpose
Input method radio	Switch between "Image upload (vision AI)" and "Paste text"
File uploader	Upload PNG/JPG/JPEG/WEBP of handwritten list (vision model reads it)
Extract rows from image button	Triggers vision AI extraction -> preview rows

Element	Purpose
Paste text area	Alternative - paste tab/comma/pipe-separated rows
Parse pasted rows button	Parse the textarea
OCR Review grid	Editable preview of extracted rows with ambiguous-match pickers
Confirm All & Stage button	Pushes all reviewed rows into pending_issues

4.3.2 Mobile camera barcode/QR scanner (expander)

Barcode / QR Scanner - opens the browser camera and reads SAP codes off labels.

Element	Purpose
Camera feed	Live preview (requires camera permission)
Manual SAP code entry text box	Fallback if camera unavailable - also accepts pasted scanner output
Detected code display	Shows the last successfully scanned code in green

When a code is scanned, the material selectbox auto-populates so you don't have to search.

4.3.3 Scan / Add New Item to Queue (main expander, expanded by default)

This is the workhorse panel. Reading top to bottom:

A. Recently-scanned ring buffer

Element	Purpose
Recent: pills	Up to 5 quick-tap buttons showing your last 5 items (SAP code + truncated description). Tapping one auto-fills the material selectbox.

B. 1. Select Material

Element	Purpose
Search by SAP Code or Description selectbox	Type-ahead search across the inventory master. Format: [SAP_Code] Description.

When you pick a material, two cards appear automatically:

C. Item Snapshot card

A compact dark card showing:

- SAP code (gold monospace)
- Material description (bold)
- 3-stat strip:
 - SITE STOCK (color-coded: red=empty, amber=below min, green=ok)
 - 30-DAY BURN (units consumed last 30 days)

- DAILY RATE
- Inline 70x22 SVG sparkline - last 30 days consumption trend
- Status badge bottom-right (OK / Low / Below Min / Empty)

D. FEFO panel (lot suggestion)

If lots exist for this item at your site:

Element	Purpose
FEFO - First Expiry, First Out header (amber)	Banner
Per-lot row	Lot number (monospace) - xqty (dim) - "Exp <date>" (amber if <90d, red if expired) - USE FIRST badge on the first row

The system will automatically attach the top-FEFO lot number to your consumption row when you submit.

E. Pull from a different lot (FEFO override) - expander

Only renders when 2 or more open lots exist for this item at your site.

Element	Purpose
Lot to pull from instead selectbox	Default: "Keep FEFO suggestion". Other options: all other open lots with expiry + remaining qty shown
Reason for override text box (200 char max)	Mandatory. Min 5 characters to activate the override
Amber confirmation banner	Appears once override is active: "FEFO override active: pulling from <chosen> instead of <suggested>"

When activated:

- Your consumption row gets Lot_Number = chosen + FEFO_Override = reason
- Audit log entry FEFO_OVERRIDE is written with full context
- A WhatsApp alert is queued to the site HOD in real time

F. 2. Fill Entry Details (dynamic form)

The form shows a field for every detail an issue entry carries. Every required field is marked with an asterisk.

Field	Type	Purpose
Date	Date picker (default: today)	When the consumption happened
Quantity	Number input (min 0.1)	How much you're issuing. Above this field is a stock badge showing current site stock; a red [!] warning appears if you type more than available.
Work_Type	Selectbox (sourced from system_settings.Work_Type)	Classification: PM Work, Breakdown, Project, Shutdown, Inspection, etc.

Field	Type	Purpose
Issued_By	Text (auto-filled with your username)	Who handed it over
Issued_To	Text (smart-defaults from your last entry)	Recipient personnel/department
Tank_No	Text	Equipment tag if applicable
Serial_No	Text (optional)	Item serial number if tracked
PR_Number	Text (smart-default)	Linked purchase request if applicable
Remarks	Text (optional)	Free-form notes

Hidden side-effects:

- Site_ID is auto-set to your site
- Lot_Number is auto-set to FEFO top suggestion (or your override)
- status is set to 'draft' until you submit

G. [!] Override Expiry Warning (amber card with checkbox)

If the system detects a short-dated batch at your site for this item, it will hard-block the Add to Grid button until you check this box, confirming you've physically pulled from the expiring batch first.

H. Add to Grid button

Validates that:

- A material is selected
- No required fields are empty
- Quantity <= current site stock (over-issue guard)
- Expiry override is checked when needed

On success: the entry is added to your staging queue as a draft, a "[OK] Added to staging queue" confirmation appears, and the page refreshes.

4.3.4 Staging Queue section

Element	Purpose
Staging Queue header + count badge	Shows how many draft rows you've accumulated
Data editor table	Edit any row inline before submitting
Save Draft Edits button	Persists edits in-place (rows stay as drafts)
Submit Grid to HOD button	Moves every draft row to awaiting-HOD-approval and queues a WhatsApp alert to the site HOD with the full item list

After submission, the rows are locked from your view - they appear in HOD Portal -> EOD Commit tab for approval.

4.4 Entry Log -> Receipt Staging

For when materials arrive at your site.

4.4.1 Upload Delivery Note (OCR) - expander

Identical lane structure to consumption OCR - upload an image of the delivery note OR paste text. Vision AI extracts rows for bulk staging.

4.4.2 Add Receipt to Queue - expander (expanded by default)

A. 1. Select Material

Element	Purpose
Search by SAP Code or Description selectbox	Same type-ahead search as consumption

When picked, a blue-tinted info card shows:

- Mat Code
- UOM

B. 2. Fill Receipt Details (dynamic form)

Field	Type	Purpose
Date	Date picker (today)	Delivery date
Quantity	Number input	Units received
Supplier	Text (optional)	Vendor name
Lot_Number	Text (optional)	If blank + Expiry_Date is set, system auto-generates LOT-YYYYMMDD-SAP
Expiry_Date	Date picker (optional)	Lot expiry - when set, triggers lot master entry
PR_Number	Text (optional)	Link to an open Purchase Request
Serial_No / Vehicle_No / DN_No / Pallet_No / Mob_From / Mob_To / Prepared_by / Driver_Name	Text (optional)	Logistics tracking fields
Remarks	Text	Free notes

C. Add to Receipt Queue button

Checks that a material is picked and every required field is filled, then adds the line to your receipt draft queue.

4.4.0 Incoming Delivery Notes from Warehouse (NEW in v3.0)

A new expander appears at the TOP of the Receipt Staging tab. It's only populated when the Warehouse has prepared a DN bound for your site AND your HOD has approved it. If empty, the expander shows:

"Nothing inbound. Logistics -> HOD-approved DNs will appear here for you to confirm physical receipt."

When DNs are inbound, each appears as its own container:

Element	Purpose
DN header line	DN <number> - PO <number> - Warehouse <id> - DN Date <date>
Line count + total qty	At-a-glance: how many SKUs, how many units total
View lines expander	Material_Code, Description, Qty, UOM, Lot_Number, Expiry_Date, Remarks - read-only preview of every line
[OK] Mark as Received button	Confirms physical receipt. One receipt is recorded per delivery line, each keeping its delivery note, warehouse and originating order so the trail is complete; the delivery closes and leaves your list. Stock figures update immediately - the Live Dashboard shows the new quantity the same minute.

When to use:

- The Warehouse delivered the materials physically to your site
- HOD already approved the DN content (you'll see it appear without any action)
- You've inspected the goods and they match the DN qty + lot

When NOT to use:

- Direct deliveries from supplier to your site (those go through the existing Add Receipt to Queue -> Submit to HOD flow below; the email-driven PR/PO path remains supported)
- Partial receipts (the current flow assumes you confirm the full DN qty - for partials, raise a Vendor Return on the diff after confirming and ask your HOD)

This is the FINAL step in the procurement chain. After you click Mark as Received:

- Logistics sees a "DN received at site" entry in their notifications
- Warehouse sees the same
- The PO and PR move toward closure if this DN completes the order

4.4.3 Receipt Draft Queue section

Element	Purpose
Receipt Draft Queue header + count badge	Number of drafts you've accumulated
Data editor table	Inline edit before submit
Save Draft Edits button	Persists
Submit to HOD for Approval button	Moves your drafts to awaiting-HOD-approval and sends the HOD a WhatsApp alert with the item list

4.5 Entry Log -> Returnable Items

For tools, gauges, fittings temporarily handed to personnel - items that should come back.

4.5.0 Smart Scan workflow (new - 2026-06)

A two-step camera flow sits at the top of the tab. Use it to skip typing borrower + tool details by hand; the existing manual form still works exactly as before and stays visible underneath.

Step 1 - Scan the employee badge

1. Click into the Smart Scan (Beta) expander.
2. Hold the borrower's printed badge in front of the camera. (Badges are generated in Admin Portal -> Employees -> Roster + Badges - see §7.6.)
3. The system decodes the QR (which carries only the ID_Number, no PII), looks it up in the Employees master, and shows:
 - [OK] Green success card - EMP-1042 - Ahmed Ali (+9665...) - the borrower's name + phone are now staged.
 - [X] Red error card - Badge couldn't be read OR the employee is inactive/suspended. Re-snap or use the manual form.

Step 2 - Scan the tool

1. After a successful badge scan, a second camera input appears. Point it at the tool.
2. The active image-recognition model (managed in Admin Portal -> Tool Catalogue, which is also where a new model is trained and promoted) classifies the image. Three outcomes:
 - Confidence ≥ 0.75 -> [OK] Auto-fill. The detected tool name + borrower details flow into the manual form below; review and click "Issue Item".
 - Confidence 0.30-0.74 -> [!] A "Top candidates" radio appears. Pick the right one and click "Use this tool". The form pre-fills.
 - No active CV model OR confidence < 0.30 -> An info banner explains the fallback. Borrower fields stay pre-filled; type the tool name into the manual form yourself.

Click "Start over (clear scan)" any time to reset and scan a different borrower.

What happens if the AI is wrong

You're never locked in. Every Smart Scan write-through goes into the manual form's fields - Name, Phone, Material name. Edit anything that's wrong before clicking "Issue Item". The submitted record records that the camera made the identification, and how confident it was, so admin reports can later show adoption and accuracy. Manually-corrected loans are not tagged as camera-identified - that's intentional, see §10 for the data model details.

Automatic WhatsApp reminders (Phase 6E)

Once a loan is in the system, the WhatsApp worker fires four reminders automatically. The cadence and recipient list is:

Offset	Severity	In-app	WhatsApp
T-2h (2 hours before due)	info	Site SK badge	Borrower
T-0 (due now)	warning	Site SK badge	Borrower
T+2h (2 hours overdue)	warning	Site SK badge	Borrower + every Site SK
T+24h (24 hours overdue)	critical	Site SK + Supervisor badges	Borrower + every Site SK + every Site Supervisor

The borrower's phone is taken first from the Employees master (for badge-scanned loans) and falls back to the phone number typed on the loan itself for older or manually-entered loans. The system remembers which reminders it has already sent for each loan, so restarting the reminder service mid-day never causes double-fires. Admins can mute any of the four events with the per-event WhatsApp toggles if a particular escalation gets too noisy.

4.5.1 Issue a Returnable Item - expander (expanded by default)

Field	Type	Purpose
Material / Tool Name	Text	What you're handing out
UOM	Text (e.g., Pcs, Set)	Unit
Quantity	Number (min 0.1)	How many
Borrower Name	Text	Person taking custody
Borrower WhatsApp Number	Text (optional, +966 ...)	For overdue alerts
Expected Return Date	Date (default: tomorrow)	When you expect it back
Expected Return Time	Selectbox: 04:15 PM, 06:15 PM, Custom Time...	Time-of-day expectation
Issue Item button	Validates name+borrower, writes to returnable_items	

4.5.2 Currently Borrowed Items section

Shows all borrowed (not yet returned) items at your site.

- Overdue red banner - when one or more items are past their expected return time
- Borrowed items table (styled HTML) with columns: ID, Material, UOM, Qty, Borrower, Phone, Given Time, Expected Return, Status (pill: green "On Loan" / red "[!] Overdue")
- Mark as Returned section:
 - Selectbox of borrowed items
 - [OK] Mark as Returned button - marks the loan returned, refreshes the figures, and confirms on screen

4.6 Entry Log -> Stock Count

For reconciling physical shelf count with system stock when they don't match (damage, expiry disposal,

miscount, found-extra).

4.6.1 1. Select Material to Count

Element	Purpose
Search by SAP Code or Description selectbox	Pick the material to reconcile
Help caption	Explains when to use this tab

4.6.2 2. Enter Count Details (only renders after picking a material)

Left column

Element	Purpose
System Qty card	Read-only dashboard showing current stock at your site (auto-fetched)
Counted Qty input	What you actually count on the shelf right now (defaults to system qty)
Variance preview banner	Color-coded: green " Found N extra" or red " Short by N" or grey "No variance"

Right column

Element	Purpose
Reason Code selectbox	9 options. Default depends on variance direction.
Notes textarea (300 char max)	Free explanation: "found behind shelf 3", "damaged in transit", etc.

Reason codes:

Code	Label
cycle_count	Cycle count correction
damaged	Damaged / unusable
expired_disposal	Expired - disposed
miscount_in	Miscount - found extra
miscount_out	Miscount - short
lost	Lost / unaccounted
theft	Suspected theft
return_to_supplier	<- Returned to supplier
other	Other (see notes)

4.6.3 Submit section

Element	Purpose
Amber warning banner	"Submitting this sends the count to your HOD for approval. No stock changes until they approve."
Submit Count for HOD Approval button	Disabled while the variance is 0. On success the count is filed as a stock adjustment awaiting HOD approval, a WhatsApp with the count details goes to the site HOD, and the submission is audit-logged.

4.6.4 Recent Adjustments at This Site (history)

Read-only table showing: ID, SAP Code, Material Name, Variance, Reason, Status, Submitted By, Submitted At.

4.7 Store Keeper - Use Cases

Use Case 1: Issue 10 Pipe Gaskets to PM Work

1. Log in -> Entry Log -> Consumption Log tab
2. Open Scan / Add New Item to Queue
3. Search "Pipe Gasket" -> pick from dropdown
4. Read the stock badge (e.g., " 156 Pcs"), confirm FEFO lot suggestion
5. Quantity: 10
6. Work_Type: PM Work
7. Issued_To: name of recipient
8. Tank_No / PR_Number: as applicable
9. Click Add to Grid
10. Repeat for any other items in this batch
11. Review the Staging Queue table -> edit if needed -> click Submit Grid to HOD

The row(s) are now in HOD's EOD queue. Stock won't decrease until HOD commits EOD.

Use Case 2: Receive a delivery of 200 units with expiry date

1. Entry Log -> Receipt Staging tab
2. Open Add Receipt to Queue
3. Pick material, fill Quantity (200), Supplier, Expiry_Date (e.g., 2027-06-11)
4. Lot_Number: leave blank (system will generate LOT-20260611-<SAP>) OR type the supplier's lot ID
5. PR_Number: link if applicable
6. Click Add to Receipt Queue
7. Click Submit to HOD for Approval when done staging

Use Case 3: Found 5 extra units of an item (physical count > system)

1. Entry Log -> Stock Count tab

2. Pick the item from the dropdown
3. Read the System Qty card (e.g., "95")
4. Counted Qty: 100
5. The variance banner turns green: " Found 5 extra"
6. Reason: Miscount - found extra
7. Notes: e.g., "Found in box behind shelf 3"
8. Click Submit Count for HOD Approval

HOD will approve in HOD Portal -> Adjustments. After approval, your live stock matches reality and a synthetic receipt row of +5 is posted to the ledger.

Use Case 4: Issue a tool to a borrower

1. Entry Log -> Returnable Items tab
2. Issue a Returnable Item
3. Material: Torque Wrench 1/2", UOM: Pcs, Qty: 1
4. Borrower Name, WhatsApp Number, Expected Return: today 06:15 PM
5. Click Issue Item

If the item is not returned by 06:15 PM, the borrower (and you) get a WhatsApp alert.

Use Case 5: FEFO override (the right bin is blocked)

1. Entry Log -> Consumption Log -> pick material with 2+ open lots
2. Scroll to FEFO panel - note the suggested lot (earliest expiry)
3. Open Pull from a different lot expander
4. Pick a different lot from the dropdown
5. Type a reason >= 5 chars (e.g., "FEFO bin blocked by pallet, clearing 1700")
6. Amber confirmation banner appears
7. Fill quantity + other fields
8. Click Add to Grid

Your HOD gets a real-time WhatsApp alert documenting the override.

4.8 Store Keeper - FAQ

Q: I can't see Live Dashboard, HOD Portal, or Reports. Did I lose access? A: No - Store Keepers only have access to Entry Log by design. If you need to view stock, ask your Supervisor or HOD.

Q: I submitted to HOD by mistake. Can I cancel? A: Not directly. Contact your HOD - they can reject the row in their EOD Commit review.

Q: The "Add to Grid" button is disabled or the form rejects my entry. A: Check three things: (a) is quantity > current stock? (b) is the expiry override needed but unchecked? (c) are all required * fields filled?

Q: I tried to issue 20 but the system says only 15 available. A: Believe the system. Either physically recount, or if there is genuinely 20 and the system disagrees, use the Stock Count tab to log a +5 adjustment.

Q: I don't have a barcode scanner. How do I quick-find items? A: Type a partial SAP code or description in the selectbox - it has type-ahead search. Or use the Recently-scanned pills if you've used the item recently.

Q: WhatsApp alerts aren't reaching the HOD when I submit. A: HOD's phone number must be set in their user profile. Tell the HOD to update it via Admin (or check with Admin directly).

Q: My borrowed item is overdue but I haven't received an alert. A: WhatsApp alerts are sent by a background service. Ask your Admin to confirm it is running - queued messages send as soon as it is back, so nothing is lost in the meantime.

Q: I uploaded an OCR image and it extracted wrong items. A: The OCR review grid lets you fix every row before staging. Pick the correct material from the suggested-matches dropdown, or type it manually.

Q: How do I see my recent activity? A: Submitted/approved items don't show on your screen. Ask HOD or Admin to filter the audit log by your username.

4.9 Printing a consumption form

Where: Execution Entries (/execution) -> Print a consumption form. Store Keeper, Supervisor and HOD.

The field fills these in by hand and photographs them; the app reads the photo. Everything about the form's design exists to make that reading reliable.

4.9.1 What is already printed, and what you write

The form prints every material for the system - name, component, SAP code and unit - so nobody ever writes a material name by hand. Reading handwritten names is the one thing the vision model is genuinely bad at, and a name that is already printed cannot be misread.

You write three things at the top - Date, Equipment / Tank No. and Area done (m²) - then, on every row, the quantity used and the lot / batch number for that material, and your name at the bottom.

[!] The lot is per row, not per form. One system draws several materials and each one arrives from a different batch, with its own certificate. Writing a single batch number at the top would be right about one material and wrong about the rest - and the certificate check at approval looks at the lot for each material, so a batch on the wrong line clears a check for material that was never used.

[!] The four small black squares in the corners are not decoration. They are what lets the app square up your photograph and show you the right row. Keep them in frame.

[!] The date box is blank on purpose. Forms are printed in batches and used today or tomorrow, so a pre-printed date would be wrong on half of them - and the date decides which day's progress your work lands on. The small date in the footer is when the blank was printed: it tells you if you are holding old paper.

[!] There are no spare rows. Only materials in the system's recipe can be recorded. Write 0 for anything you did not use; never add a material by hand, because there is no box for it and the app cannot match it.

4.9.2 [!] Rows that look almost identical

Some systems list the same product several times as separate components. LSC8 prints "Cumicrete PU MF 300 - 3mm" four times - as Comp-A, Comp-B, Comp-C and Comp-D, each with its own SAP code.

They are different materials with different quantities. Check the component letter, not just the product name.

4.9.3 The QR code

The square in the top-right holds the site, the system, the sub-activity, this sheet's own number, and which page of the form it is. It is read by a scanner, not by the AI, which is why none of those things can be got wrong.

[!] Photograph the whole page, including the QR. A photo that crops it out cannot be matched to anything, and you will be asked to retake it.

[!] A long form has more than one page, and each page has its own code. Photograph and upload every page. Upload them in any order - they all join the same entry. Until they are all in, the missing pages' materials read 0, and the app tells you which page it is still waiting for.

Do not submit an entry that is still waiting for a page. Those materials would be recorded as unused, and the figures would look entirely reasonable.

4.9.4 [!] Every download is a new sheet

Downloading the form twice gives you two different sheets with two different numbers - not two copies of one form.

That is deliberate. The app has to be able to tell "you printed a second sheet" from "you photographed the same sheet twice", and it does that by the number on the paper. If you print a spare, it is a genuinely separate form.

An HOD can see every form printed and not yet filed, and who printed it.

4.9.5 Printing several at once - and [!] never photocopying one

Set Forms to the number you need and press Download. You get one PDF containing that many forms, each with its own number and its own QR code, and each printed SHEET 3 OF 50 at the top so you can sort the pile and see if one is missing.

[!] Never photocopy a form. This is the most important sentence on this page. A photocopy repeats the QR code, and the QR code is how the app knows which sheet it is looking at. Two sheets carrying one code cannot be told apart: the second one you send in will either be refused as already filed, or worse, filed against the first one's tank. If you need fifty, print fifty here.

[!] The number is FORMS, not pieces of paper. A system with more than 18 materials prints as several pages - that is still one form, with one QR across its pages. The line under the box does the arithmetic for you and changes as you type:

> 50 forms x 2 pages = 100 A4 sheets

Check that line before you send it to a printer.

You can print up to 200 forms in one go. Above that, print in runs: a download that fails halfway leaves the app holding records for paper that never reached you, and those show as outstanding sheets until somebody clears them.

4.9.6 If the materials change after you print

If someone edits the system's recipe after your form was printed - adds a material, reorders them, changes a code - the app will refuse the photo and ask for a fresh form.

It is not being awkward. The app matches your handwriting to materials by row position, so a sheet printed against different rows would file your quantities against the wrong materials - and the numbers would look perfectly reasonable. Changing a material's rate is fine and does not invalidate printed paper.

4.9a Surface Shield material issued from the store - where its area is recorded

Phase 13.

The gap this closes. When a Surface Shield material is issued from the general Inventory, the ledger records that a drum left the shelf. It does not record what that drum covered - which system, which vessel, how many square metres. So the estimator could not see it and nobody could compare it against the recipe.

Every such issue now appears in a queue until somebody says what it was for.

4.9a.1 What the Store Keeper does

Nothing new, and one optional extra.

You already pick the Lining System Code before issuing a Surface Shields material - that has not changed. Beside it there is now an Equipment / tank box, filtered to the equipment that system actually applies to.

[!] Leave it blank if you do not know. You are handing over a drum; where it ends up is often decided in the plant. A guess here becomes a wrong attribution later, and the queue will simply ask. A blank is not a failure.

[!] You are never asked for the area. You cannot know it - the drum leaves before it is applied. That question belongs to the person who applied it.

4.9a.2 What the field does - the queue

Where: Execution Entries -> the Surface Shield consumption queue. Store Keeper, Supervisor and HOD.

The queue lists every Surface Shield consumption with no area recorded against it, oldest first - except rows the HOD rejected, which sit at the very top (see §4.9a.6). For each one you supply three things:

System Code	Pre-selected when the store keeper already chose one. Otherwise pick from the list - it offers only the systems whose recipe actually contains that material.
Equipment / tank	Filtered to the equipment carrying the system code you picked.
Area covered (m ²)	What that material actually covered.

[!] The system code is asked, never guessed. If the app cannot tell which system a draw was for, it offers you the candidates rather than choosing one. A guessed code compares the draw against the wrong benchmark - and a wrong benchmark looks finished, where a blank looks unfinished. Unfinished is safer.

[!] The queue reaches back. It is not an inbox that starts today: it holds every unattributed Surface Shield consumption, including issues from months ago and rows brought in by an Excel import. Expect it to be long the first time you open it, and work oldest first.

[!] Consumption filed from a printed form is NOT in this queue, and must not be. A photographed consumption form already recorded its system, its equipment and its area, and the area has already been credited. Adding it here would count the same drum against the same vessel twice.

4.9a.3 What happens to the numbers

Nothing is deducted. The stock left the shelf when it was issued; this records what it was for, not that it went.

Each submission is compared against the recipe: rate x area is what the system expected, and the difference is the variance. Every row then goes to the HOD - see §4.9a.4.

> [!] Your estimator figures do not move. Readiness, completion, achievable > area and the buy list are calculated from the Material Estimator's own > workbook figures and are untouched by any of this. What you gain is > visibility of what was actually drawn, beside the plan.

4.9a.4 [!] Every row goes to the HOD - the +/-10% band decides the order

Every Surface Shield consumption is reviewed, whatever the variance. There is no band inside which a row files itself.

What the +/-10% tolerance does is set priority. A draw more than 10% above or below the benchmark is flagged High Priority and sits at the top of the HOD's queue; the rest follow, oldest first.

[!] A variance that cannot be calculated is High Priority too. If the recipe has no line for that material in that system, there is no benchmark to compare against - and "we cannot work this out" is exactly the thing worth looking at. It is not treated as zero.

An admin can change the band. Doing so re-orders the queue and changes nothing about rows already filed - each one keeps the reading it was measured against at the time.

4.9a.5 [!] When the Excel workbook changes a row that was already assigned

The Excel sync now updates a row in place when you edit it in the workbook. It never adds a second

copy of the same consumption, and it never deletes or detaches the system code, equipment and area already recorded against it.

If the edit changes the quantity, date or tank, the area recorded against that consumption was measured against figures that no longer exist. So:

1. The row comes back to the queue marked Edited in Excel, showing the new quantity with the old one struck through.
2. Press Re-assign. The previous system code, equipment and area are filled in - confirm them or correct them.
3. The HOD is notified and decides.
4. If the HOD had already approved the old figures, those keep counting until the new ones are approved. On approval the original entry is updated with the new figures, and the equipment's completed area moves by the difference only - it is never counted twice.
5. If the HOD rejects the re-assignment, the approved figures stay and the row returns to the queue with the HOD's reason.

If the edit changes nothing that the assignment depends on - or you push the same workbook again - nothing is asked again. Editing Remarks, Received by or Prepared by never sends a row back.

[!] Changing a row's Date in the workbook makes it a different row. The sync keeps the old one and reports it as vanished rather than deleting it, because a deleted row could take somebody's work with it. An administrator can remove vanished rows with the sync's --prune-vanished option, which only deletes rows nothing is attached to.

4.9a.6 [!] When the HOD rejects an assignment - it comes back to you

A rejection is not the end. The row returns to the top of the queue for the person who filed it to correct.

What the HOD does. Press Reject and write the reason. A reason is required: the field cannot correct something nobody told them was wrong.

What the field sees.

1. The row is back in Needs an area, sorted above everything else, with a red Rejected - Needs Correction badge and the HOD's reason written beside it.
2. The person who filed it also gets a notification with the reason.
3. Press Correct & resubmit. The answer that was rejected is filled in, with the reason shown above it. Change the system code, equipment or area.
4. Press Resubmit to HOD. The row leaves your queue and goes straight back to the HOD, marked Resubmitted after rejection, with their earlier reason beside it. The HOD is notified.

Nothing counts until the HOD approves: a rejected assignment credits no area to any equipment and adds nothing to the Estimator's consumed figure.

A re-assignment of an already-approved row that the HOD rejects comes back the same way; the approved figures keep counting while you correct it.

4.10 Filing a consumption form

Where: Execution Entries (/execution).

This replaced the old store-keeper-first flow on 27 August 2026. The record now starts with the paper you filled in.

You fill the form -> Store Keeper verifies -> HOD approves

4.10.1 Step 1 - photograph it

Upload the photo under Upload a filled form. JPG, PNG, HEIC or PDF.

[!] Photograph the whole page, including the QR code. The QR is what tells the app which form this is - a photo without it cannot be matched to anything and will be refused.

[!] Only forms printed from "Print a consumption form" belong here. A handwritten consumables sheet or a supplier's delivery note has no QR code and never will. Those are read in Entry -> OCR Import instead - and if you upload one here by mistake, the refusal will say so and link you to the right page.

Reading a page takes minutes, not seconds, and the card now tells you how many. You will see a live counter and the usual time for that kind of page: roughly 6 1/2 minutes for a printed consumption form, 3 1/2 for a 30-row handwritten sheet, 1 1/2 for a delivery note. A dense or badly-lit page takes longer and the card says so rather than pretending to be nearly finished.

You can leave the page; the read carries on and the result will be waiting.

> [!] If it says "This read was interrupted", it means the server process doing > the reading stopped - not that your page was bad. Waiting longer will not > help; press Read it again. This is the honest version of a problem that > used to look like the page simply never finishing: before 2026-09-02 the card > claimed "usually takes under a minute" and then showed a spinner with no > elapsed time, so a perfectly good six-minute read and a dead worker looked > identical, and people quite reasonably gave up on both.

4.10.2 Step 2 - check every figure

The app opens a draft with what it read. Beside each row is the crop of your photograph it read that number from, so you are checking against the paper and not against memory.

[!] Where the handwriting was not certain, the box is left EMPTY rather than guessed. Those rows are listed at the top and marked in gold with what was actually written. Type the real number in. A guessed figure would post straight to stock with nobody questioning it.

Then correct anything else, pick the equipment from the list if it was not read, and give the two reasons - material and manpower - which are required on every entry, even one with no variance at all.

4.10.3 Step 3 - the store keeper verifies

It goes to the store keeper, not to the HOD. They check your quantities against what actually left the shelf and may change them - every change costs a written reason, shows to the HOD in red, and you are told about it before the entry is approved rather than after.

Blasting and buffing skip this step: there is no material to verify.

4.10.4 Step 4 - the HOD approves

[!] Approval is what deducts the material and posts the area. Nothing before it moves a quantity, which is what makes correcting a figure safe at every earlier step.

The HOD sees the whole chain on each row:

Colour	What it is
grey	what the camera read
amber	what you filed
red	what the store keeper corrected
purple	what the HOD settled on

A row everybody agreed on shows one number and the word "agreed". The colours only appear where something actually changed - which is what makes a red one worth looking at.

[!] Rejection is final. A rejected entry cannot be revived; raise a new one from a fresh form.

4.10.5 [!] Store keepers: stop raising a separate issue

The execution entry is now the only way lining material leaves the ledger. Raising a material issue for the same drum as well would deduct it twice - and nothing would show it until somebody counted the shelf.

4.10.6 [!] When a certificate is missing

If a Surface Shield material has no test certificate or no quality clearance, the HOD's screen says so before they press approve, naming the lines.

The material has already been applied by then, so this is a paperwork gap rather than something anyone can prevent. The HOD may approve anyway - the button says "Approve WITHOUT clearance" and will not work until a reason is typed - and the Head of Qualities is notified every time.

Often the simpler fix is a corrected lot number: the check runs after the HOD's edits, so fixing the batch can clear it with no override at all.

4.10.7 Things the app will refuse, and why

The same sheet twice. Each printed form is filed once. Two people photographing one form, or one person retrying on a bad signal, produce different files of identical paper - the number on the sheet is the only thing that can tell.

A form printed before the materials changed. If someone edits the system's recipe after your sheet was printed, the rows no longer line up. The app matches your handwriting to materials by row position, so your quantities would be filed against the wrong materials and would look entirely reasonable. Print a fresh form and copy the figures across.

A form printed for another site.

A photo it could not read. It says so rather than creating a blank entry - a blank entry that gets submitted is a consumption of zero, recorded silently.

10. Data Model & Concept Reference

10.1 Core movement tables

Table	What it stores	Identity-math role
inventory	Master catalogue: SAP_Code (the unique identifier), Description, Material_Code, UOM, Minimum_Qty, Unit_Cost	The "items that exist" - defines what can be moved
receipts	Every received unit (post-commit). Includes Lot_Number, Expiry_Date, Supplier, PR_Number, Unit_Cost	+ adds to stock
consumption	Every consumed unit (post-EOD-commit). Includes Lot_Number, FEFO_Override, Work_Type	- subtracts from stock
returns	Tools and equipment returned to inventory	+ adds back
pending_issues	Pre-commit staging for consumption (status: draft -> pending_hod -> approved/rejected -> committed)	- does NOT affect stock
pending_receipts	Pre-commit staging for receipts	- does NOT affect stock

Identity formula: $\text{Current Stock} = \text{Total Received} - \text{Total Consumed} - \text{Total Returned}$. It is never stored as a number anyone can type over; it is recalculated from the ledger every time it is shown.

10.2 Document-type tables

Table	What it stores	Lifecycle
stock_adjustments	Physical-count reconciliations	pending_hod -> approved (posts synthetic ledger row) / rejected
requests	Cross-site material transfers	pending -> approved / rejected -> fulfilled
pr_master	Purchase Request lines (manual or PDF-extracted)	status: open / closed; workflow_state: draft -> submitted -> approved -> in_progress -> received
returnable_items	Tools temporarily issued out	borrowed -> returned / overdue
lots	Lot master metadata (FEFO source-of-truth)	open -> exhausted / expired / disposed / quarantine

10.3 Supporting tables

Table	Purpose
users	Auth + role + site + phone
pending_users	Self-registration queue
system_audit_log	Immutable activity record
system_settings	Dropdown values (Work_Type, Site)
app_settings	Key/value config (thresholds, maintenance_mode, last_backup_at)
whatsapp_queue	Outbound message queue
pwa_tokens	Access tokens for the offline-capable mobile app
bug_reports	User-submitted issues/ideas
report_schedules	Scheduled report definitions
report_archive	Generated report metadata

10.4a Procurement chain tables (NEW in v3.0)

Table	What it stores	State machine
warehouses	Master of receiving locations	active / inactive
vendors	Supplier master	active / inactive
purchase_orders	PO header (PO_Number UNIQUE)	open -> partially_delivered -> delivered -> closed / force_closed / cancelled
po_items	PO line items with rl_bl_family tag	open -> partially_delivered -> delivered / returned / closed / force_closed
po_shipment_schedule	Parsed PO Annexure rows	pending / shipped / delivered / delayed / cancelled
po_assignments	Logistics -> Warehouse routing	assigned -> acknowledged -> partial / received / closed / cancelled
delivery_notes	DN header (DN_Number UNIQUE)	DN state machine: draft -> pending_logistics -> pending_hod -> pending_sk -> received (or rejected from any pending)
dn_items	DN line items	pending / received / partial / returned / cancelled
po_returns	Vendor returns (raised by any role)	open -> vendor_acknowledged / resupplied / cancelled

Table	What it stores	State machine
po_reschedule_requests	Date-change asks	pending -> approved / rejected
po_force_closures	Force-closure audit log	(terminal - write-once)
app_notifications	In-app bell inbox	Unread -> read, stamped with the time it was read
delivery_reminders_sent	Record of which T-2 / T-1 / T-0 reminders have gone out	Write-once - the same reminder can never be recorded twice

10.4 Views

View	Purpose
v_live_stock	Per-SAP global stock (sum across all sites)
v_site_stock	Per-(SAP, Site) live stock - the canonical "what's at this site"
v_expiring_stock	Lots/receipts in expiry buckets
v_supplier_activity	Per-supplier receipt rollups
v_lot_balance	Per-lot Received_Qty / Consumed_Qty / Remaining_Qty (identity math)

11. Status Codes, Reason Codes & Glossary

11.1 Common status pills

Status	Color	Meaning
pending	grey	Awaiting decision
flagged	amber	Has a concern (e.g., zero stock at site)
approved	green	Approved, ready for commit
rejected	red	Soft-rejected (row preserved for audit)
committed	gold	Posted to permanent ledger
draft	grey	User can still edit
submitted	blue	Sent forward; awaiting next stage
in_progress	amber	Active (e.g., PR with supplier)
received	green	Fully delivered
open	blue	Active, has pending balance
closed	green	Fully fulfilled
OK / Low / Below Min / Empty	green/amber/orange/red	Stock vs minimum status
Expired / Critical / Warning	red/red/amber	Expiry buckets
sent	green	WhatsApp delivered

11.2 Adjustment reason codes

Code	Label	When to use
cycle_count	Cycle count correction	Routine periodic count
damaged	Damaged / unusable	Physical damage
expired_disposal	Expired - disposed	Expired and physically discarded
miscount_in	Miscount - found extra	Found more than system shows
miscount_out	Miscount - short	Found less than system shows
lost	Lost / unaccounted	Genuinely missing
theft	Suspected theft	Deliberate removal
return_to_supplier	<- Returned to supplier	Quality issue, sent back
other	Other (see notes)	Anything else (must explain in notes)

11.3 PR workflow states

- draft - manual creation, not yet sent
- submitted - sent to procurement
- approved - procurement greenlit
- in_progress - supplier engaged
- received - fully received

11.4 Lot statuses

- open - has remaining quantity, available for FEFO
- exhausted - fully consumed (auto)
- expired - past expiry date (manual or background)
- disposed - physically removed (manual)
- quarantine - held pending inspection (manual)

11.4a DN states (NEW v3.0)

draft -> pending_logistics -> logistics_approved -> pending_hod -> hod_approved -> pending_sk -> received With rejected as terminal from any pending state.

11.4b PO + PO line states (NEW v3.0)

PO header: open -> partially_delivered -> delivered -> closed / force_closed / cancelled PO item line: open -> partially_delivered -> delivered -> returned / closed / force_closed

11.4c Force-closure target types (NEW v3.0)

Code	Label
pr	Whole PR closed
po	Whole PO closed
po_item	Single line on a PO closed

11.4d Reschedule + vendor return states (NEW v3.0)

Reschedule: pending -> approved / rejected Vendor return: open -> vendor_acknowledged / resupplied / cancelled

11.4e RL/BL family tags (NEW v3.0)

Tag	Meaning	Detection rule
RL	Rubber Lining	The Material Code or Description contains "RL-", "RUBBER LINING" or "RUBBER-LINING"
BL	Brick Lining	It contains "BL-", "BRICK LINING", "BRICK-LINING" or "BRICK MATERIAL"
(none)	Neither family	The default when nothing above matches

The system works this out for you. An item is never tagged as both - if the wording matches both families, RL wins.

11.4f Notification severity (NEW v3.0)

Severity	Visual	When used
info (blue)	Info pings - PR submitted, PO issued, assignments	
warning (amber)	T-2/T-1 reminders, reschedule requests, rejections	
success (green)	DN approved successfully, delivery completed	
critical (red)	T-0 reminder, force-closures, urgent escalation	

11.4g Logistics-status on PR rows (NEW v3.0)

Code	Meaning
site_draft	HOD has the PR but hasn't submitted to Logistics yet
submitted	Sitting in Logistics queue waiting for PO issuance
in_po	A PO has been issued against this PR line
closed	PR fulfilled normally
force_closed	Logistics force-closed the PR with a reason

11.5 Glossary

Term	Meaning
FEFO	First-Expiry-First-Out - issue oldest-expiring lot first
EOD	End-of-Day commit (HOD action that finalizes the day's

Term	Meaning
	transactions)
PR	Purchase Request
DN	Delivery Note (logistics document)
OCR	Optical Character Recognition - used for bulk-staging from images
PWA	Progressive Web App - the offline mobile companion
RBAC	Role-Based Access Control
WAL	A database setting that lets many people read while one person writes. Nothing to configure - it is on
Standard cost	Per-item Unit_Cost on inventory master (vs weighted-average cost which would require receipts-history)
Identity math	Stock derived from movements, never stored as a counter

12. FAQ - Master Index by Role

12.1 General (any role)

Q: I forgot my password. A: Contact your Admin. Admin -> Access Control -> Force Password Reset.

Q: I get "Permission denied" when clicking a page. A: Your role doesn't have access. See §2.2 for the access matrix.

Q: I want to suggest a feature or report a bug. A: Any page -> sidebar -> Bug/Feature reporting dialog. Admin reviews in Admin Portal -> Reports & Bugs.

Q: The page is blank or broken. A: Hard-refresh (Cmd+Shift+R / Ctrl+F5). If it persists, check with Admin - they can see audit logs to diagnose.

12.2 Store Keeper - see §4.8

12.3 Supervisor - see §5.5

12.4 HOD - see §6.14

12.5 Admin - see §7.13

21. 2026-08 Feature Update - What Changed

Everything in this chapter is live. Grouped by what you would notice.

21.1 Reports and exports look the same everywhere now

Every Excel export - not just the Material Estimator's - now carries the branded layout: the GI logo, a "Report Generated / Generated By" band, a title bar and a framed, filterable table with the header frozen. Previously only three SME workbooks looked like that and everything else came out as a bare grid.

> If you have automation reading these files: the header row is now row 6 > and data begins on row 7 (rows 1-4 are the logo and meta band, row 5 the > title bar). Anything that assumed a header on row 1 needs adjusting.

PDF reports no longer overlap or cut off text. Columns are now measured against their actual content and share the page fairly, and long values wrap onto more lines instead of being truncated. Previously every column got an equal slice of the page regardless of need, so a long material description was drawn on top of the next column while Date and UOM wasted most of theirs - and anything past 24 characters was silently deleted.

QR badge and bin labels were fixed the same way: a long employee name used to run past the edge of its sticker onto the one beside it.

21.2 The Material Estimator understands your order data correctly

Ordered_Qty is the TOTAL quantity procured for the project, and Available_Qty is the part of it that has already arrived - available is a subset of ordered, not a separate pile beside it.

The estimator used to add the two together, which double-counted every unit already on the shelf. On real data that understated the buy list by 22,951 units across 22 of 30 materials, and on GI-8005763 - where all 143,000 ordered units had arrived - it read 286,000 and reported nothing to buy against a demand of 152,685, hiding a 9,685-unit shortage completely.

What you will see in the UI:

- Second-tier quantities now read "Pending Delivery" - the part of the order still on its way - never just "Ordered".
- Second-tier coverage reads "When delivered".
- Reports carry a Total Procured column beside them.

Feasibility is unchanged and still judges physical stock only: a tank cannot be built with a purchase order.

21.3 A view-only Auditor role

See §20.

21.4 You are signed out after 30 minutes of inactivity

Leave the app untouched for 30 minutes and it signs you out and returns you to the login screen. A warning appears two minutes before with a button to stay signed in, and any real activity - a click, a keystroke, a scroll - resets the clock silently.

This matters most on shared site terminals, where an unattended session was previously good until the browser was closed.

21.5 Search anything with CmdK

Press CmdK (Mac) or Ctrl-K (Windows) anywhere in the app.

The palette already jumped to pages; it now also searches your inventory live. Type a SAP code, part of a material name or a material code and the matching items appear under a Materials heading - pick one and you land on its Material Intelligence card. Results are scoped to what your role may see.

21.6 The Hub Assistant is faster and more accurate

The assistant used to be handed its entire allowed portion of this manual on every single question. For an Admin that was the whole ~180 KB document, which is slow to process; for everyone else it was the first 800 characters of each chapter, which is the wrong 800 whenever the answer is further down.

It now finds the passages that actually match your question and reads only those - a 97.7 % smaller prompt for Admins, and noticeably better answers for everyone, because the relevant paragraph is now present in full instead of being cut off.

A parsing bug was fixed at the same time: numbered comment lines inside the worked examples in the Operations chapter were being mistaken for chapter headings, which silently replaced chapters 1-4 - Introduction, Roles & Permissions, Login, and the Store Keeper Manual - with fragments of a restore procedure. Ask "how do I log in?" today and you get §3.1, the login screen.

The role filter is unchanged and still applied before retrieval: a Store Keeper's context physically cannot contain an Admin chapter.

21.7 Serialised tool and equipment tracking

Two hammers share one item code, so scanning either label used to resolve to the same inventory line - the system could not tell you which one was in your hand or where the other had gone. Assets solves that.

Each physical item gets its own record, identified by its serial number. A scan now resolves three ways, and the order is deliberate:

What the label carries	What you get
A serial number or asset tag	That exact item, immediately - there is nothing to choose

What the label carries	What you get
Only the item code, and several units exist	A "which one is in your hand?" list to pick from
Only the item code, and no units are registered	The ordinary material view - the system does not invent an asset

Recording where something is. Open the item, tap Move, and either pick a rack or describe the place in words - not everything lives on a shelf, and "loaded on truck 4771" is a real answer. If the device can supply coordinates, they are captured alongside the update, never in front of it: a declined permission or a yard with no signal still records the move, with the coordinates simply absent. A recorded position becomes a map link on the item.

Status covers both what condition a tool is in - working, not in use, under repair - and where it sits in the stores workflow. The full history of every move is kept and is never overwritten, so "where has this been" is a question with an answer.

> Coordinate capture needs a secure connection. Over plain HTTP the browser > refuses to supply a position - the move still saves, without coordinates. If > your team is on the installed app or the company web address, this is already > the case.

> Your spreadsheet seeds this; the app owns it. A location typed in the > Consumption Log creates the asset on the next sync. From then on, what somebody > recorded in the app wins: a re-sync will not move a tool, revert its status, or > erase a captured position.

21.8 The warehouse rack locator

"Which rack is this in?" now has an answer. Search by material name, by code, or by scanning a label, and the system tells you the shelf to walk to - ordered so that the first result is where to go first, even when a material is stocked in three places.

- A material you stock but nobody has located comes back marked not located, rather than being left out. Silence would read as "we do not stock it", which sends somebody out to buy what is already on a shelf.
- Scanning the rack itself answers the reverse question - everything that is meant to be on that shelf. That turns a stock count from a hunt into a checklist.
- The locator is built into the CmdK search, so it is two keystrokes from anywhere in the application.

21.9 The Material Estimator handles messy tank references

The Consumption Log records a tank reference typed by hand, and the same vessel appears written four different ways. The system now recognises those as one vessel automatically - that alone accounted for 58 of 103 rows.

Where a reference is genuinely ambiguous, it refuses to guess. TNK-091 exists on both TRAIN J and TRAIN K; either answer would look completely plausible in every report afterwards, so those rows wait in a queue for someone who knows. Resolve the reference once on the Actual Consumption tab and every row waiting on it is tagged.

21.10 For administrators - local tooling

Two maintenance scripts ship with the system, documented in the administrator runbook: one puts the database and the public connection to sleep and wakes them again for a laptop that is not always on, and one takes and restores database snapshots, with an option to run nightly at 02:00.

> The previous nightly backup had been failing silently for 25 consecutive > nights - it pointed at a script that a repository reorganisation had moved. > Anyone relying on it had no local backups at all. The replacement is verified > by actually restoring into a scratch database on every change, which is the > only check that means anything.

21.11 Downloaded files now protect you from your own spreadsheet

If you type something that begins with an equals sign, a plus, a minus or an at-sign into a text field - a remark, a note, a reason - you may notice a small apostrophe in front of it when you open the downloaded Excel or CSV file.

That is deliberate, and nothing has been lost.

Spreadsheet programs treat a value beginning with one of those symbols as a calculation to run, not as words to show. That is fine when you write the formula yourself. It is not fine when the text arrived from somewhere else: a note typed at one site, exported by a manager and opened on their computer, would be run by their spreadsheet without anyone choosing to. The apostrophe tells the spreadsheet to show the text exactly as it was written instead.

What this means in practice:

- Numbers are never touched. Quantities, prices, areas and every total in every report stay real numbers, and every total still adds up.
- A plain number that happens to be negative, such as -5, is left exactly as it is - it is a number, not a calculation.
- Ordinary text - material names, supplier names, ordinary notes - is untouched.
- If you want the plain text back in your own copy, delete the apostrophe in that cell.

You are most likely to see this on remarks and reason fields, because those are the places people type freely. It applies to every download the system produces: reports, master-data exports and the Material Estimator workbooks alike.

21.12 Phase 9 - what changed for you

Five things changed at once, and one of them changes how you work every day.

21.12.1 [!] Consumption is filed from paper now

This replaces the store-keeper-first flow entirely. Full detail in §4.10.

You fill a printed form -> Store Keeper verifies -> HOD approves

- Print a form per lining system (§4.9). It lists every material for that system, so nobody writes a material name by hand - that is the one thing the reader is genuinely bad at.
- Fill in the date, equipment and area at the top, and a quantity and a lot number on every row.
- Photograph the whole page including the QR code and upload it.
- Check every figure. Where the handwriting was not certain the box is left empty rather than guessed, and those rows are marked.
- The store keeper verifies your quantities next, not the HOD. Their corrections show to the HOD in red, and you are told before it is approved.

[!] Approval now deducts the material as well as posting the area. Store keepers: stop raising a separate material issue for lining work - the execution entry is the only writer now, and doing both deducts the same drum twice.

[!] Rejection is final. Raise a new entry from a fresh form.

[!] Each printed sheet is used once, and a sheet printed before its system's materials changed will be refused. Both rules exist because your handwriting is matched to materials by row position.

21.12.2 WBS numbers, at last

The WBS # column was blank on every entry in the system - not because the feature was missing, but because there was no screen to reach it. There is one now: HOD -> WBS & Work Types (§16.5).

- Add your WBS numbers, then map each work type to one.
- An issue that leaves the WBS box blank gets the number its work type maps to.
- Work types are now a fixed list you curate, which is what stops civil and Civil becoming two cost centres.
- Nothing is enforced until you add the first row. Turning it on is your act.
- WBS is applied forward only; historical entries are not restamped.

21.12.3 A new chart: Efficiency by Day

Man-Hours -> Efficiency by Day (§19.6). For one lining system, how much manpower went into Equipment A against Equipment B - normalised on man-hours per m², which is the only figure that survives a change of size.

[!] The line is the running figure, not the day's, and it breaks on a day with hours and no area. Both are deliberate; §19.6 explains why.

21.12.4 The Manpower Planner tells you what nights actually buy

Three changes to the arithmetic:

- Nights buy calendar time, not a smaller payroll. The total headcount is unchanged - nobody works both shifts - but the plan now shows how many days the night crew saves.
- The day/night split follows your roster, not a division by two. This site runs about 20 on days against 80 on nights; halving the requirement understated the night crew fourfold.
- Hiring idle trades no longer flatters the plan: capacity counts only the roles the job needs.

21.12.5 "Labor" is now "Manpower"

The page, the menu entry and every heading now say Manpower - the word everyone here actually uses.

> If you have automation reading the API: the field names did not > change. Done_SQM_Labor and Labor_Variance_Pct are still spelled exactly > as they were. Only what you see on screen moved, so nothing you have built > against those keys needs touching.

21.13 Photo reading - the two failures that are fixed

If you tried to read a photographed Daily Consumption log or the new printed consumption form and it did not work, this is what was happening and what has changed. Nothing about how you use it is different - you still photograph the page and wait.

21.13.1 "The log came back empty, or said it could not be read"

The reader was being cut off part-way through a long sheet. It read your page correctly, got as far as about row 13 or 14 of a 30-row log, and ran out of room - and because the answer was incomplete, the system threw away all of it, including the thirteen rows it had read perfectly.

This is also why Delivery Notes always worked: a note is four or five items and always finished comfortably. It was never that the reader could not handle your handwriting or the free-form table.

Now: it is given enough room for a full 30-row sheet, and if a page is ever still too long, the rows it did read are kept and only the unfinished one is dropped. On the operator's own test photo this went from nothing at all to all 30 rows.

21.13.2 "The new PDF just hangs, then fails"

A full page of handwriting genuinely takes the reader several minutes. The system was giving up after four - often within seconds of it finishing - and then telling you "the vision model is not reachable", which sent people off to check a service that was running perfectly.

Now: it waits long enough for a real page, and if something genuinely is wrong it says which of the two it is: still working, or actually unreachable.

21.13.3 Two smaller things you may notice

- A quantity the reader could not make out now arrives BLANK, not as 0. It used to show a confident zero, which reads like a number somebody wrote and nobody re-checks. A blank box is a question, and the row will not submit until you answer it.
- A photo that produced nothing now says so. Previously a failed read could finish looking merely "empty", which is indistinguishable from a photo of a blank form. If the page could not be read, you are told to retake it.

21.13.4 If a photo still will not read

Nothing has changed about the advice, but it is worth repeating: get the whole page in frame, keep the lighting even, and hold the camera square to the paper. The Paste tab works with the AI switched off

entirely and is always available. Typing the entry in by hand is never wrong.

21.13.5 [OK] Ditto marks now come through (2026-09-02)

Where: Entry -> OCR Import (/entry/ocr), the Consumption log lane.

If you have been retyping the Name, Tank No. and Product Name columns after every photo read, that was a real fault and it is fixed.

Most people write a ditto mark - " or ? - instead of repeating themselves down a column, and on a typical sheet that is most of the page. The reader was seeing those marks perfectly well and quietly returning nothing for them, because to a reader "same as above" is not a word to transcribe. The step that was supposed to copy the value down was looking for the mark, found an empty cell instead, and left it empty. On the sheet this was diagnosed against, that was 19 of 30 tank numbers, 14 of 30 names and 8 of 30 product names.

Now: a dittoed cell is filled in from the row above, whether the reader returns the mark or returns nothing.

Two things worth knowing about how it behaves:

- A cell that was filled in for you is marked [?]. If the reader gave us nothing and we worked the value out from the row above, the row carries an information marker saying so. A cell where you actually wrote a ditto mark is not marked, because that one is what the paper says rather than what we concluded. It never blocks anything - it just lets you see the difference.
- Rows you never wrote on stay empty. The S.No. column is printed on all thirty rows whether or not anybody used them, so if you filled in ten rows, rows 11 to 30 come back blank. They are not filled in with the last person's name.

Nothing about how you use the page changed: photograph it, press Validate against the spec, review, stage.

22. Quality, Safety, Employees & Procurement (QSEP)

Everything in this chapter went live in August 2026. It adds four things that used to live on paper or in somebody's memory: who checked the material, who is wearing the safety gear, where a person works, and what the scanned purchase paperwork actually said.

Read section 22.1 first even if only one of the four concerns you - the QC approval rule changes what happens when a Store Keeper tries to issue certain materials, and that affects the whole site.

22.1 Quality Control - the new role and the issue block

What a QC does

A Quality Control inspector checks material that carries risk if it is wrong. Today that means the Surface Shields category - 36 materials out of the 466 in the master list. Nothing else in the catalogue is affected by any rule in this section.

> ### The rule, in one line > > Material without an MTC CAN be sent and dispatched to site. It CANNOT be > issued or consumed at the site. > > Receiving it is fine. Putting it on a Delivery Note and driving it to site is > fine. Handing it to a worker is not. If you have been told otherwise, that > came from a document written before 2026-08-12 - the rule below is what the > system actually does.

There are two separate gates. Both apply at the same moment - the instant before material goes to a worker - but they are satisfied by different people, which is what makes them worth telling apart:

Gate	When it applies	What it demands	Who is stopped	Who fixes it
Material Test Certificate	When a Store Keeper issues material to the field	A certificate on file covering that material at that site	The Store Keeper at the issue form	Logistics (or the warehouse, or the SK)
QC approval	The same moment	A QC must have inspected and approved that quantity at that site	The Store Keeper at the issue form	The QC

Material may be received, and may travel to site, with neither. That is deliberate and was decided by the operator. Holding a truck at the gate for paperwork that is sitting in somebody's inbox costs more than it saves - and worse, refusing to record a delivery does not stop it arriving; it only stops the system knowing about it, so real material sits in a yard invisible to the stock report and to everyone planning around it. What material may not do is reach a worker's hands unchecked. So both gates bind at issue.

When a controlled item is received without a certificate, Logistics is notified automatically to chase it from the supplier. Nobody has to remember.

You should almost never upload the same certificate twice

The certificate follows the material down the chain. Whoever has it files it once:

- Logistics attaches it to the purchase order line - before the goods have even shipped, and without knowing which site will end up with them;
- the warehouse attaches it to the Delivery Note as the material leaves;
- the Store Keeper uploads it at the site, from the Receive form or from the banner on the Issue form.

Any one of those clears the material for issue at the destination site, and the issue form tells you which document cleared it and where it came from. A certificate filed for one site does not clear another - it attests to one batch from one production run.

If you find yourself uploading a second copy of a document the company already has, stop and report it. That is a fault, not a workflow.

Raising an inspection

An inspection is opened automatically when controlled material is received. The QC does not create it by hand. It appears in Quality -> Inspections with a status of pending.

Deciding an inspection

Open the inspection and enter the quantity you approve. The status follows from the number, not from a separate choice:

You approve	Status becomes
Nothing (zero)	Rejected
Less than the submitted quantity	Partially approved
The whole submitted quantity	Approved

Rejected material stays in stock. It is not automatically sent back to the vendor and it does not disappear. It sits where it is, unusable, and is blocked from issue until somebody decides what happens to it. That was an explicit instruction: an automatic vendor return removes the evidence before anyone has looked at it.

What the inspector sees, and the Return No (added August 2026)

The inspection screen now shows the material's name, with the SAP and material codes underneath it. Before this it showed only the codes, which meant judging a material by a number rather than by what it is.

If a Material Test Certificate is on file, the certificate number appears with an Open certificate link that opens the actual document - in the queue and again in the inspection dialog. Previously it only said that a certificate existed, with no way to read it, which meant approvals were being made against documents nobody had opened.

When an inspector rejects any quantity, the system issues a Return No such as QCR-20260813-41. It is announced to the inspector on the spot, listed against the inspection, and sent to both the site's Store Keeper and its Head of Department. The Store Keeper types it into Return Stock and the return form fills itself in - see §13.10.

> This does not change the rule above. The Return No is an invitation for a > person to raise a return; nothing moves, and nothing is sent back to a vendor, > until the Store Keeper posts it and the HOD approves it.

The daily missing-certificate alert (added August 2026)

Every morning the system checks for Surface Shield material that is in stock with no Material Test Certificate on file, and tells the people who can do something about it:

Where the material is	Who is told
In a warehouse	Logistics, the Warehouse User, and the warehouse's QC
At a site	That site's Store Keeper, HOD and QC, and Logistics

Logistics appears on both lists because they are the only people who can obtain the document from the supplier. One message per location lists everything outstanding there, rather than one message per material.

The alert repeats every morning until the certificate is uploaded, and then stops on its own. That repetition is intended: this is a standing condition, not a one-off event, and the material cannot be issued to anybody while it lasts.

What the Store Keeper sees when the block fires

The issue form refuses with a message that names the actual numbers, so nobody has to guess which of three situations they are in:

- No inspection at all - "no quality inspection exists for it at this site. It cannot be issued to the field until a QC has checked the material and its certificate."
- Inspected but nothing approved yet - names how many inspections are still pending.
- Partly approved and used up - "QC has approved 40 and 40 is already issued or staged, leaving 0 - not enough for 10. Ask the QC to inspect the remaining stock."

The clearance is pooled per material per site, not per individual pallet. What is counted against it is everything already issued or staged awaiting HOD approval, from the date of the first inspection onward - issues that predate quality control entirely are not counted, or the site would be frozen forever by its own history.

> This does not change the FEFO or over-issue behaviour. Those remain > warn-and-record, as they always have. The quality block is about whether the > material is fit to use, which is a different question from whether the shelf > count agrees.

Creating a QC account

Quality -> QC Accounts, available to an HOD, a Warehouse User or Logistics. You create a QC inside your own scope: an HOD creates one for their site, a Warehouse User for their warehouse. Set a site or a warehouse, never both.

Moving a QC to another site is a request, not an action - the HOD raises it and an Admin decides. A QC whose remit could be changed by the person whose work they inspect is not an independent check.

22.2 PPE - issued through the ordinary form

There is no separate "Issue PPE" page, and this is deliberate. Safety boots leave the store the same way a drum of resin does: through Entry Log -> Issue. The form recognises that the item is PPE and asks for the extra details on the spot.

Select a PPE item and three fields appear:

Field	When it is required	Why
Employee ID number	Always	PPE is tracked against a person, not a site
Safety Approval document	Unless a usable-time rule says otherwise	Somebody signed for this issue
Reason for early replacement	Only if the worker's current item has not expired yet	A boot replaced after three weeks is either a defect or a leak

The stock movement and the PPE record are written together. There is no second ledger and no separate PPE stock - the quantity leaves through the normal issue path, so stock levels, burn rate, reports and every other number stay correct without knowing anything about PPE.

The messages you will see

- "...is PPE - name the employee receiving it (their ID number), so it can be tracked against them."
- "worker 30816 is not in the employee master" - the person must exist on the roster first.
- "worker 30816 is at site RIYADH, not CNCEC - transfer them first if they have moved" - see 22.3.
- "Ahmed already holds this item, issued 2026-07-02 and good until 2026-10-02 - give a reason for replacing it early."

Usable time

Safety & People -> PPE Usable Time (Store Keeper or HOD) is where you say how long an item lasts: 90 days for gloves, 365 for a helmet, and so on. A rule can be global or set for one site, and the site's rule wins where both exist.

> [!] Until a rule exists for an item, two things follow. The issue will > always demand a Safety Approval document, and the item gets no expiry date > - which means it will never appear in the forecast below. If the PPE Forecast > looks empty, this is almost always why.

An expiry is a suggested replacement date, not a restriction. Nothing stops a worker using gear past it, no alert is sent to anybody's phone, and the worker is not blocked from anything.

The 15-day forecast

Safety & People -> PPE Forecast lists PPE expiring in the next fifteen days and tells you what to order:

> suggested = expiring - what is on the shelf - what is already on order

If the answer is zero because thirty units are already on an open purchase order, that is the correct answer and not a broken screen. The list shows the names of the people whose gear is expiring,

because a column of quantities cannot be sanity-checked by a human and a column of names can. A 90-day issue rate is shown beside the suggestion as a cross-check, never folded into it.

22.3 Employees - one person, one number

The employee ID number is the person. It is unique across the whole company, not per site, and everything else hangs off it.

Safety & People -> Employees shows the roster. An HOD can transfer someone to another site, and the transfer takes effect immediately - no approval step, because a person who has physically moved has already moved.

PPE history follows the person, not the site. Transfer a worker from one site to another and the boots they are wearing are still on their record. This falls out of keying on the ID number rather than being a feature bolted on top, which is why it cannot quietly stop working.

Admins get a timeline for any employee: every site they have worked at, with dates, and everything they currently hold. Use it when somebody asks "who had this, and when".

Employees -> Data Quality lists records that cannot be used and says why - a missing ID number, a duplicate, an inactive worker still holding gear. Check it before a PPE rollout rather than discovering the gaps one refused issue at a time.

22.4 Procurement - drafted delivery notes, urgency, and scanned paperwork

Delivery Notes draft themselves

When a warehouse receives goods against a purchase order, the system now prepares the Delivery Notes the sites are waiting for, grouped so that R/L and B/L material never share a note.

They are created as drafts. A person still adds the vehicle and driver and submits them, because the system cannot know that a truck exists. A notification tells the warehouse the drafts are waiting.

If a draft cannot be created - an over-shipment, a missing certificate - the goods receipt still stands. The stock genuinely arrived; losing that record because a convenience failed would be far worse. The reason is written to the audit log and the clerk cuts the note by hand.

Auto-drafting can be switched off in Settings.

Assigning a purchase order to a warehouse (changed August 2026)

A purchase order goes to one warehouse. The Purchase Orders list now shows a Warehouse column naming where each order was sent, and once an order has been assigned the Assign button is replaced by an assigned tag.

Previously the button stayed on every row for ever, and the list gave no indication that an order had already been routed. Pressing it a second time created a second assignment and told a second warehouse to expect the same goods, with nothing anywhere reporting a problem.

- Assigning the same warehouse again - a double-click, or an old tab - is accepted and changes nothing. No duplicate, no second notification.
- Assigning a different warehouse is refused, and the message names the warehouse that already holds the order. Re-routing an order another warehouse is already expecting is a decision for a person to make, not something the system will do quietly.

Shipping a delivery note (changed August 2026)

Pressing Ship now asks for two things before the delivery leaves:

- The delivery note number printed on the physical document - the carrier's or the supplier's number. This is not the DN number the system generated; the point is to connect the two, so that a delivery can be reconciled later without a telephone call.
- A scan or a photograph of the signed note. The camera button opens the phone camera directly, which is the practical option at a loading bay.

Both are required. If either is missing you are told which one, separately - one is typing and the other is scanning, and being sent to redo the half you had already done is its own small waste.

A Material Test Certificate is still optional here and works exactly as before. A certificate covers the material and is inherited from the purchase order or the delivery note down to the receiving site, so nobody uploads it twice. A delivery note covers this shipment and is inherited by nothing. They are different documents answering different questions.

Once shipped, the document number and a download link appear beside the delivery for everyone who touches it - the warehouse, Logistics, the HOD, Quality and the receiving Store Keeper. The Store Keeper is the one holding the paper copy while the truck waits, so this is where it matters most.

> Deliveries shipped before this change show "not shipped yet" in that > column. There is no document for them because none was ever captured, and > inventing a number would be worse than saying so.

Marking a reschedule urgent

A delivery reschedule can be marked urgent. An urgent one goes out immediately instead of waiting for the 4 p.m. evening digest. Use it sparingly - a channel where everything is urgent is a channel nobody reads.

Reading a scanned PR or PO

Upload the supplier's paperwork and the system reads the lines off it, so nobody retypes forty rows.

The file is stored before it is read, and this matters more than it sounds: the document that defeats the reader is precisely the one somebody will need to look at later. A failed read still leaves you the document.

There are two ways a document gets read, and the system chooses by whether the file actually contains text - not by whether it is a PDF:

The file	What happens
A PDF exported from a system	Text is read directly. Fast, exact.

The file	What happens
A PDF that is a photograph or a scan of a signed page	Read by the vision model. Takes longer; you poll for the result.

A printed-signed-and-scanned purchase order contains no text at all, only pictures of text. Before this change such a file was reported as read successfully with zero lines found - which is worse than an error, because nothing on screen distinguished "this order has no items" from "I could not read a word of it".

Once read, confirming the document links it to the purchase request or order you created, so the original scan is one click from the record. Offering a purchase order scan to a purchase request is refused by name - the two are not interchangeable.

> A quantity the reader could not make out is left blank, not zero. A zero on > a purchase order looks like an answer and would be ordered as one.

22.5 Assets - one hammer, one row, anywhere in the company

A serialised asset is now unique by its part number and serial number across the entire company, not per site. The same hammer cannot be registered twice by two sites who both think they have it.

Moving an asset between sites is a request approved by the HOD of the site it is leaving - the site giving something up is the one that needs to agree. On approval the asset's site changes and its old rack assignment is cleared, since a shelf in one yard means nothing in another. The move is recorded as a movement, so "where has this been" still answers correctly for the leg between the two sites.

Trying to re-register a serial that exists elsewhere is refused with a message that says where it actually is and what to do about it.

22.6 Passwords

The minimum password length is 8 characters, and a password must now also contain an uppercase letter, a number and a special character. This replaced a flat 12-character minimum with no composition requirement.

The rule is applied identically everywhere a password is set - admin creation, password reset and self-registration. Self-registration previously enforced a much weaker rule than the admin screens did.

The Request Access page now offers Quality Control in its role list.

26. Practice Mode - learn the system without touching Live

Practice is a separate copy of GI Hub for training. It has every page, every role and every workflow Live has, but it runs on its own made-up data in its own database. Nothing you do in Practice can reach Live: not a receipt, not an approval, not a message.

Use it to learn a job before doing it for real, to rehearse something you do rarely (a return, a stock count, a PR), or to try something out without worrying about the result.

26.1 Switching between Live and Practice

On the sign-in screen, above the username box, there is a Live | Practice switch.

- Live is selected unless you choose otherwise. It is the real system.
- Choose Practice and the page reloads, ready to sign in to Practice.
- Choose Live again to go back.

Things to know about switching:

- Switching signs you out. You sign in again in the environment you picked. This is deliberate: a browser is in one environment at a time.
- Your Live password does not work in Practice, and a Practice password does not work in Live. They are separate systems with separate accounts (§26.3).
- The installed apps (Windows, macOS, Android, iOS) have the same switch.

26.2 How you know you are in Practice

The page tells you in four ways, and none of them can be dismissed:

Where	What you see
Top of every page	An amber bar: PRACTICE - practice data, nothing here is real.
Page header	An amber PRACTICE tag beside the title, and an amber line under the header that stays when you scroll
Browser tab	The tab title starts with PRACTICE -
Anything you print	A large diagonal PRACTICE - NOT REAL DATA watermark

These come from the server you are connected to, not from the switch. The server reports which environment it is, and the page shows what it says.

[!] If you see a red bar saying "Environment mismatch", the switch and the server disagree. Everything you try to save will be refused until that is fixed. Sign out, pick the environment again, and tell your

administrator if the red bar comes back.

26.3 The practice accounts

Practice has one shared account per role. Your trainer gives you the password for these accounts.

Role	Practice account
Store Keeper	practice.storekeeper
Supervisor	practice.supervisor
Head of Department	practice.hod
Quality Control	practice.qc
Head of Qualities	practice.qchod
Warehouse	practice.warehouse
Logistics	practice.logistics
Auditor (view-only)	practice.auditor
Admin	practice.admin (separate password, held by your trainer or administrator)

The site-based accounts work at site CNCEC, and the Warehouse account works at warehouse WH-01.

Everyone in a class uses the same accounts, so you will see each other's practice entries. That is normal.

26.4 What is in Practice

The data is made up but realistic. The categories, units of measure, lining systems, work types and tank numbers are real. The material names, SAP codes, people, vendors, quantities, prices and dates are all invented. Nothing in Practice is a real person or a real stock figure.

It is the same data the training videos use. If a video shows Tutorial Demo Gasket Set (SAP 899001), you can find it in Practice.

Work is already waiting so every role has something to do: receipts, issues and a return waiting for HOD approval, a draft purchase request, and one already with Logistics.

26.5 What works differently in Practice

Everything works exactly as in Live, except these four things. Each one is switched off on purpose:

In Practice...	Why
No WhatsApp message or email is ever sent. Notifications still appear in the bell.	A practice alert must never reach a real phone or inbox.
Photo reading (OCR) is switched off. The page explains	The AI engine is shared with Live and takes minutes per

In Practice...	Why
this, and you can still paste text or type rows in.	page, so trainees must never queue ahead of real forms.
Training videos watched in Practice are not counted. You can watch as much as you like, but I have watched and understood this is refused.	Training records are kept in Live only. Sign in to Live to have a video counted.
Two-factor authentication is off, and cannot be turned on.	The accounts are shared, so one person's authenticator would lock everyone else out.

26.6 Offline entries stay where they were made

If you lose signal while saving an entry, GI Hub keeps it and sends it when the connection returns (§3.7). In Practice this still works, with one extra guarantee: an entry saved in Practice is only ever sent to Practice. If you switch to Live before it has been sent, it waits, and it is sent the next time this browser is back in Practice. It can never be sent to Live, even if you sign in to Live on the same device.

26.7 Resetting Practice (admins)

Over time Practice fills up with everyone's practice work. An admin can put it back to its starting state:

1. Sign in to Practice as practice.admin.
2. Open Admin Console -> Practice. The page shows when the data was built and when, and by whom, it was last reset.
3. Click Reset Practice data... and type RESET PRACTICE DATA to confirm.

The reset takes about a second. It removes everything anyone has done in Practice, and everyone signed in to Practice is signed out. The approval queues are filled up again. Live is not affected in any way. It is a different database, and the Practice server has no permission to open it.

The Practice tab and the reset only exist in Practice. In Live there is no such button, and the server has no such function.

How administrators set up and run Practice on a server is in §17.9.